

Slide 1: KidMood: Classroom Emotion Recognition

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- ITAI 1378 - Final Project**
- Tier 2: Transfer Learning**

Slide 2: Problem & Motivation

- Teachers cannot monitor every student's emotional state at all times.**
- Early signs of sadness, fear, or anger may be missed.**
- KidMood is an assistive awareness tool, not a diagnosis system.**

Slide 3: Solution Overview

- **Input: face image or frame**
- **Model predicts: Angry, Fear, Happy, or Sad**
- **Output: emotion label + confidence score**

Slide 4: Technical Approach

- Task: Image Classification**
- Model: MobileNetV2 transfer learning**
- Framework: TensorFlow/Keras**
- Evaluation: accuracy, macro F1-score, confusion matrix**

Slide 5: Model Architecture

- 48x48 grayscale FER image
- Resize to 224x224 + convert to RGB
- MobileNetV2 pretrained backbone
- Dense + Dropout layers
- 4-class Softmax output

Slide 6: Dataset & Preprocessing

- Dataset: FER-2013**
- Project classes: Angry, Fear, Happy, Sad**
- Preprocessing: normalize, resize, grayscale-to-RGB**
- Split: train / validation / test**

Slide 7: Live Demo

- **Run 2-3 test images through inference script**
- **Show predicted label and confidence**
- **Show output image saved to results/images**
- **Backup: recorded demo video**

Slide 8: Results

- Metrics will be generated after training
- Accuracy + macro F1-score
- Confusion matrix
- Success and failure examples

Slide 9: Success & Failure Cases

- **Success: clear frontal expression images**
- **Hard case: low-resolution or ambiguous expression**
- **FER-2013 labels may be noisy or subjective**

Slide 10: Key Learnings

- Transfer learning makes training practical**
- Emotion classes can be visually similar**
- Evaluation matters more than perfect accuracy**

Slide 11: Future Work

- Add webcam demo**
- Add face detection/cropping**
- Try ResNet50 comparison**
- Build mood trend dashboard**

Slide 12: Acknowledgments

- Instructor guidance
- FER-2013 dataset
- TensorFlow/Keras pretrained models
- AI assistance documented in docs/AI_usage_log.md